Linear Algebra II HW 7

1. Let V be an inner product space (not necessary finite dimension) and $T:V\to V$ be a linear transformation such that

$$\langle T(\boldsymbol{u}),T(\boldsymbol{u})\rangle=\langle \boldsymbol{u},\boldsymbol{u}\rangle \text{ for all } \boldsymbol{u}\in V$$

Show that $\langle T(\boldsymbol{u}),T(\boldsymbol{v})\rangle=\langle \boldsymbol{u},\boldsymbol{v}\rangle$ for all $\boldsymbol{u},\boldsymbol{v}\in V$

Solution

For any vectors $\boldsymbol{u},\boldsymbol{v}\in V,$ we have

$$\langle T(\boldsymbol{u}+\boldsymbol{v}),T(\boldsymbol{u}+\boldsymbol{v})\rangle=\langle \boldsymbol{u}+\boldsymbol{v},\boldsymbol{u}+\boldsymbol{v}\rangle$$
$$\langle T(\boldsymbol{u})+T(\boldsymbol{v}),T(\boldsymbol{u})+T(\boldsymbol{v})\rangle=\langle \boldsymbol{u}+\boldsymbol{v},\boldsymbol{u}+\boldsymbol{v}\rangle$$
$$\langle T(\boldsymbol{u}),T(\boldsymbol{u})\rangle+2\langle T(\boldsymbol{u}),T(\boldsymbol{v})\rangle+\langle T(\boldsymbol{v}),T(\boldsymbol{v})\rangle=\langle \boldsymbol{u},\boldsymbol{u}\rangle+2\langle \boldsymbol{u},\boldsymbol{v}\rangle+\langle \boldsymbol{v},\boldsymbol{v}\rangle$$
$$\langle \boldsymbol{u},\boldsymbol{u}\rangle+2\langle T(\boldsymbol{u}),T(\boldsymbol{v})\rangle+\langle \boldsymbol{v},\boldsymbol{v}\rangle=\langle \boldsymbol{u},\boldsymbol{u}\rangle+2\langle \boldsymbol{u},\boldsymbol{v}\rangle+\langle \boldsymbol{v},\boldsymbol{v}\rangle$$
$$\langle T(\boldsymbol{u}),T(\boldsymbol{v})\rangle=\langle \boldsymbol{u},\boldsymbol{v}\rangle$$

2. Let V be an inner product space (not necessary finite dimension) and $T:V\to V$ be a mapping such that

$$\langle T(\boldsymbol{u}),\boldsymbol{v}\rangle=\langle \boldsymbol{u},T(\boldsymbol{v})\rangle \text{ for all } \boldsymbol{u},\boldsymbol{v}\in V$$

(a) Show that if $\langle \boldsymbol{u}_1,\boldsymbol{v}\rangle=\langle \boldsymbol{u}_2,\boldsymbol{v}\rangle$ for all $\boldsymbol{v}\in V,$ then $\boldsymbol{u}_1=\boldsymbol{u}_2.$

(b) Show that T is a linear transformation.

(c) Show that $\mathrm{Ker}(T)=\mathrm{Im}(T)^{\perp}$ and $\mathrm{Ker}(T)=\mathrm{Ker}(T^2).$

Solution

(a) Suppose $\langle \boldsymbol{u}_1,\boldsymbol{v}\rangle=\langle \boldsymbol{u}_2,\boldsymbol{v}\rangle$ for all $\boldsymbol{v}\in V.$ Then $\langle \boldsymbol{u}_1-\boldsymbol{u}_2,\boldsymbol{v}\rangle=0$ for all $\boldsymbol{v}\in V.$ Choose $\boldsymbol{v}=\boldsymbol{u}_1-\boldsymbol{u}_2.$ We have $\|\boldsymbol{u}_1-\boldsymbol{u}_2\|^2=\langle \boldsymbol{u}_1-\boldsymbol{u}_2,\boldsymbol{u}_1-\boldsymbol{u}_2\rangle=0.$ Hence $\boldsymbol{u}_1=\boldsymbol{u}_2.$

(b) Let $\boldsymbol{u}_1,\boldsymbol{u}_2$ be any vector in V and a, b be any scalar in $\boldsymbol{R}$. Then

$$\begin{aligned}\langle T(a\boldsymbol{u}_1 + b\boldsymbol{u}_2), \boldsymbol{v}\rangle &= \langle a\boldsymbol{u}_1 + b\boldsymbol{u}_2, T(\boldsymbol{v})\rangle \\ &= a\langle \boldsymbol{u}_1, T(\boldsymbol{v})\rangle + b\langle \boldsymbol{u}_2, T(\boldsymbol{v})\rangle \\ &= a\langle T(\boldsymbol{u}_1), \boldsymbol{v}\rangle + b\langle T(\boldsymbol{u}_2), \boldsymbol{v}\rangle \\ &= \langle aT(\boldsymbol{u}_1) + bT(\boldsymbol{u}_2), \boldsymbol{v}\rangle \quad \text{for any } \boldsymbol{v} \in V.\end{aligned}$$

By (a), we have

$$T(a\boldsymbol{u}_1 + b\boldsymbol{u}_2) = aT(\boldsymbol{u}_1) + bT(\boldsymbol{u}_2)$$

for any vectors $\boldsymbol{u}_1, \boldsymbol{u}_2 \in V$ and any scalar a, b in $\boldsymbol{R}$.

(c) To show $\text{Ker}(T) \subset \text{Im}(T)^{\perp}$.

Let $\boldsymbol{u} \in \text{Ker}(T)$. $\langle \boldsymbol{u}, T(\boldsymbol{v})\rangle = \langle T(\boldsymbol{u}), \boldsymbol{v}\rangle = 0$ for all $T(\boldsymbol{v}) \in \text{Im}(T)$. That means $\boldsymbol{u} \in \text{Im}(T)^{\perp}$.

To show $\text{Im}(T)^{\perp} \subset \text{Ker}(T)$.

Let $\boldsymbol{u} \in \text{Im}(T)^{\perp}$. Then $\langle \boldsymbol{u}, T(\boldsymbol{v})\rangle = 0$ for all $\boldsymbol{v} \in V$. We have $\langle T(\boldsymbol{u}), \boldsymbol{v}\rangle = 0 = \langle \boldsymbol{0}, \boldsymbol{v}\rangle$ for all $\boldsymbol{v} \in V$.

Hence $T(\boldsymbol{u}) = \boldsymbol{0}$ by (a). That is $\boldsymbol{u} \in \text{Ker}(T)$.

To show $\text{Ker}(T) \subset \text{Ker}(T^2)$.

Let $\boldsymbol{u} \in \text{Ker}(T)$. That means $T(\boldsymbol{u}) = \boldsymbol{0}$. We have $T^2(\boldsymbol{u}) = T(T(\boldsymbol{u})) = T(\boldsymbol{0}) = \boldsymbol{0}$. Hence $\boldsymbol{u} \in \text{Ker}(T^2)$.

To show $\text{Ker}(T^2) \subset \text{Ker}(T)$.

Let $\boldsymbol{u} \in \text{Ker}(T^2)$. Then $\langle T(\boldsymbol{u}), T(\boldsymbol{u})\rangle = \langle \boldsymbol{u}, T^2(\boldsymbol{u})\rangle = 0$. Hence $T(\boldsymbol{u}) = \boldsymbol{0}$ and $\boldsymbol{u} \in \text{Ker}(T)$.

3. Let $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ form an orthonormal basis for $\boldsymbol{R}^2$ and let $\boldsymbol{u}$ be a unit vector in $\boldsymbol{R}^2$. If $\boldsymbol{u} \cdot \boldsymbol{u}_1 = \frac{1}{2}$, determine the value of $|\boldsymbol{u} \cdot \boldsymbol{u}_2|$.

Solution

Since $\boldsymbol{u}$ be a unit vector in $\boldsymbol{R}^2$, $\boldsymbol{u} = a_1\boldsymbol{u}_1 + a_2\boldsymbol{u}_2$, $a_1 = \boldsymbol{u} \cdot \boldsymbol{u}_1 = \frac{1}{2}$, $a_2 = \boldsymbol{u} \cdot \boldsymbol{u}_2$ and $a_1^2 + a_2^2 = 1$.

$$|\boldsymbol{u} \cdot \boldsymbol{u}_2| = |a_2| = \frac{\sqrt{3}}{2}$$

4. Let $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ be an orthonormal basis for an inner product space V and let

$$\boldsymbol{u} = \boldsymbol{u}_1 + 2\boldsymbol{u}_2 + 2\boldsymbol{u}_3 \quad \text{and} \quad \boldsymbol{v} = \boldsymbol{u}_1 + 7\boldsymbol{u}_3.$$

Determine the value of each of the following:

(a) $\langle \boldsymbol{u}, \boldsymbol{v} \rangle$

(b) $\|\boldsymbol{u}\|$ and $\|\boldsymbol{v}\|$

(c) The angle θ between $\boldsymbol{u}$ and $\boldsymbol{v}$

Solution

(a) $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 1(1) + 2(0) + 2(7) = 15$

(b) $\|\boldsymbol{u}\| = \sqrt{1^2 + 2^2 + 2^2} = 3$ and $\|\boldsymbol{v}\| = \sqrt{1^2 + 7^2} = \sqrt{50}$

(c) $\cos\theta = \dfrac{15}{3\sqrt{50}} = \dfrac{1}{\sqrt{2}}, \quad \theta = 45^\circ$

5. The set

$$S = \left\{ \tfrac{1}{\sqrt{2}}, \cos x, \cos 2x, \cos 3x, \cos 4x \right\}$$

is an orthonormal set of vectors in $C[-\pi, \pi]$ with inner product defined by Theorem 5.4.5.

(a) Use trigonometric identities to write the function $\sin^4 x$ as a linear combination of elements of S.

(b) Use part (a) and Theorem 5.5.7 to find the values of the following integrals:

(i) $\int_{-\pi}^{\pi} \sin^4 x \cos x dx$

(ii) $\int_{-\pi}^{\pi} \sin^4 x \cos 2x dx$

(iii) $\int_{-\pi}^{\pi} \sin^4 x \cos 3x dx$

(iv) $\int_{-\pi}^{\pi} \sin^4 x \cos 4x dx$

Solution

(a) $\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2$

$$= \frac{1-2\cos 2x+\cos^2 2x}{4}$$

$$= \frac{1-2\cos 2x+\frac{1}{2}(1+\cos 4x)}{4}$$

$$= \frac{3}{4\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$$

(b) $L = \pi$

(i) $\langle 1, \cos x \rangle = \langle \cos 2x, \cos x \rangle = \langle \cos 4x, \cos x \rangle = 0$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^4 x \cos x dx = \frac{3}{8}\langle 1, \cos x \rangle - \frac{1}{2}\langle \cos 2x, \cos x \rangle + \frac{1}{8}\langle \cos 4x, \cos x \rangle = 0$$

$$\int_{-\pi}^{\pi} \sin^4 x \cos x dx = 0$$

(ii) $\langle 1, \cos 2x \rangle = \langle \cos 4x, \cos 2x \rangle = 0 \quad \langle \cos 2x, \cos 2x \rangle = 1$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^4 x \cos 2x dx = \frac{3}{8}\langle 1, \cos 2x \rangle - \frac{1}{2}\langle \cos 2x, \cos 2x \rangle + \frac{1}{8}\langle \cos 4x, \cos 2x \rangle = -\frac{1}{2}$$

$$\int_{-\pi}^{\pi} \sin^4 x \cos 2x dx = -\frac{\pi}{2}$$

(iii) $\langle 1, \cos 3x \rangle = \langle \cos 2x, \cos 3x \rangle = \langle \cos 4x, \cos 3x \rangle = 0$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^4 x \cos 3x dx = \frac{3}{8}\langle 1, \cos 3x \rangle - \frac{1}{2}\langle \cos 2x, \cos 3x \rangle + \frac{1}{8}\langle \cos 4x, \cos 3x \rangle = 0$$

$$\int_{-\pi}^{\pi} \sin^4 x \cos 3x dx = 0$$

(iv) $\langle 1, \cos 4x \rangle = \langle \cos 2x, \cos 4x \rangle = 0 \quad \langle \cos 4x, \cos 4x \rangle = 1$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^4 x \cos 4x dx = \frac{3}{8}\langle 1, \cos 4x \rangle - \frac{1}{2}\langle \cos 2x, \cos 4x \rangle + \frac{1}{8}\langle \cos 4x, \cos 4x \rangle = \frac{1}{8}$$

$$\int_{-\pi}^{\pi} \sin^4 x \cos 4x dx = \frac{\pi}{8}$$